Student ID:

Lab 05 Task

Your company, Intians Assemble, is expanding its headquarters and requires a new network design to support various departments. The IA headquarters consists of the following departments across multiple floors: Engineering - Floor 1, Marketing - Floor 2, Sales - Floor 3, Administration - Floor 4, Management - Floor 5

Suppose they all share the prefix **192.(i).(i+3 % 14)**. Below is the required devices for each of the departments:

$$
\begin{aligned}
E(i) &= (7i + 3) \bmod 41 + 10 && \text{(Engineering)}\\
M(i) &= (5i + 11) \bmod 26 + 5 && \text{(Marketing)}\\
S(i) &= (11i + 7) \bmod 36 + 5 && \text{(Sales)}\\
A(i) &= (3i + 17) \bmod 18 + 3 && \text{(Administration)}\\
G(i) &= (13i + 1) \bmod 9 + 2 && \text{(Management)}
\end{aligned}
$$

where i is the last two digits your student ID
Now, by completing the following table with the most efficient design decision, design the topology with correct network flow, for each network attach at least 1 end devices.

Subnetting Worksheet:

Department	Devices	Network	Mask	Usable Host Range
Engineering	$E(i)$	________	/____	________
Marketing	$M(i)$	________	/____	________
Sales	$S(i)$	________	/____	________
Administration	$A(i)$	________	/____	________
Management	$G(i)$	________	/____	________

here, fix i as 53. Now do proper subnetting and explain me the whole process

LAB 05 – VLSM Subnetting (i = 53)

I'll do everything step-by-step and then give you a ready table you can copy.

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1) Compute each department's device count

Given

- **E(i) = (7i + 3) mod 41 + 10**
- **M(i) = (5i + 11) mod 26 + 5**
- **S(i) = (11i + 7) mod 36 + 5**
- **A(i) = (3i + 17) mod 18 + 3**
- **G(i) = (13i + 1) mod 9 + 2**

For **i = 53**:

- Engineering **E = 15**
- Marketing **M = 21**
- Sales **S = 19**

- Administration **A = 17**
- Management **G = 8**

(Each value already includes just "devices"; the network needs ≥ that many usable host IPs.)

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2) Base network (shared prefix)

They "share the prefix" **192.(i).(i+3 % 14)**. The notation usually means **(i+3) mod 14**.

- i = 53 → (53 + 3) mod 14 = **0**, so the **base /24** is **192.53.0.0/24**.

  If your instructor meant *i + (3 % 14)* (i.e., +3 without mod), the base would be **192.53.56.0/24**. In that case, just replace the third octet "0" with "56" in the table below—everything else stays the same.

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3) Pick the smallest subnet for each department (VLSM)

We need the smallest block where **2^h − 2 ≥ devices**.

- 21 → needs **/27** (32 addrs, 30 usable)
- 19 → **/27**
- 17 → **/27**
- 15 → **/27** (since /28 gives only 14 usable)
- 8 → **/28** (16 addrs, 14 usable)

So we need **four /27s** and **one /28** inside the /24.

We allocate **largest first** to avoid fragmentation. /27 blocks step by 32 in the last octet; /28 blocks step by 16.

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4) Assign subnets (from low to high)

Starting from **192.53.0.0/24**:

- **Marketing (21)** → **192.53.0.0/27**

    - Network: 192.53.0.0
    - Usable: **192.53.0.1 – 192.53.0.30**
    - Broadcast: 192.53.0.31
- **Sales (19)** → **192.53.32.0/27**
    - Usable: **192.53.32.1 – 192.53.32.30**
    - Broadcast: 192.53.32.31
- **Administration (17)** → **192.53.64.0/27**
    - Usable: **192.53.64.1 – 192.53.64.30**
    - Broadcast: 192.53.64.31
- **Engineering (15)** → **192.53.96.0/27**
    - Usable: **192.53.96.1 – 192.53.96.30**
    - Broadcast: 192.53.96.31
- **Management (8)** → **192.53.128.0/28**
    - Usable: **192.53.128.1 – 192.53.128.14**
    - Broadcast: 192.53.128.15

(Plenty of room remains in the /24 for future growth.)

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5) The requested worksheet (copy-paste)

Subnetting Worksheet (Base: 192.53.0.0/24)

Department	Devices	Network	Mask	Usable Host Range
Engineering	15	192.53.96.0/27	255.255.255.224	192.53.96.1 – 192.53.96.30
Marketing	21	192.53.0.0/27	255.255.255.224	192.53.0.1 – 192.53.0.30
Sales	19	192.53.32.0/27	255.255.255.224	192.53.32.1 – 192.53.32.30
Administration	17	192.53.64.0/27	255.255.255.224	192.53.64.1 – 192.53.64.30
Management	8	192.53.128.0/28	255.255.255.240	192.53.128.1 – 192.53.128.14

If your class wants “gateway IP”, you can pick the first usable (e.g., 192.53.0.1) or the last usable (e.g., 192.53.0.30) as a convention—just keep it consistent across all subnets.

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6) Quick explanation of the process (so you can do any i)

1. **Find device counts** using the given formulas.
2. **Fix the base /24** from the prefix 192.(i).((i+3) mod 14).
3. **Sort departments by required hosts (desc)**.
4. For each, pick the **smallest subnet** where **2^h − 2 ≥ hosts** → this gives the **CIDR** and **mask**.
5. **Lay subnets left-to-right** inside the /24: /27 blocks jump by 32 in the last octet; /28 by 16.
6. For each block, note **network**, **usable range**, **broadcast**.
7. (Optional) Reserve a consistent **gateway IP**.

May Allah put barakah in your studies—clear steps, clean subnets, no collisions. If you want, I can also sketch a simple topology (core router + L3 VLAN interfaces for each subnet) next.

c) Consider a router that interconnects three subnets: Subnet 1, Subnet 2, and Subnet 3. Suppose all of the interfaces in each of these three subnets are required to have the prefix 223.1.17 with subnet mask 24. Also suppose that Subnet 1 is required to support at least 60 interfaces, Subnet 2 is to support at least 90 interfaces, and Subnet 3 is to support at least 12 interfaces. Provide three network addresses (of the form a.b.c.d/x) that satisfy these constraints.

Solution:

subnet 1 - 223.1.17.128/26
subnet 2 - 223.1.17.0/25
subnet 3 - 223.1.17.192/28

Rubric:

- 2 points for each correct network address

now explain me fully how this is solved

Here’s how that solution is built—slow, clean VLSM style. 🧮

1) What we're given

- All subnets must live inside the **same /24 block**: **223.1.17.0/24** (i.e., addresses 223.1.17.0–223.1.17.255)
- Minimum usable hosts (interfaces):
  - Subnet 1 → **60**
  - Subnet 2 → **90**
  - Subnet 3 → **12**

Quick host→prefix rule

For a subnet with prefix /x, the total addresses = 2^(32−x).
Usable = total − 2 (network & broadcast).
Pick the **smallest** block where usable ≥ need.

2) Pick the right sizes (CIDR) for each need

- Need **90** → smallest block with ≥90 usable:
  - /26: 64 total → 62 usable ❌ (too small)
  - **/25: 128 total → 126 usable** ✅
- Need **60** → smallest block with ≥60 usable:
  - **/26: 64 total → 62 usable** ✅
- Need **12** → smallest block with ≥12 usable:
  - /29: 8 total → 6 usable ❌
  - **/28: 16 total → 14 usable** ✅

So we require: **one /25**, **one /26**, **one /28** inside the /24.

> Tip: Always allocate **largest first** to avoid fragmentation.

3) Carve the /24 using alignment (block sizes)

Inside **223.1.17.0/24**:

- A **/25** is a block of **128** addresses → starts at multiples of 128:

  - **223.1.17.0/25** covers **.0 – .127**
- A **/26** is a block of **64** addresses → starts at multiples of 64:
  - Next free aligned start after .127 is **.128** → **223.1.17.128/26** covers **.128 – .191**
- A **/28** is a block of **16** addresses → starts at multiples of 16:
  - Next free aligned start after .191 is **.192** → **223.1.17.192/28** covers **.192 – .207**

Everything fits, nothing overlaps, and we're still inside the /24. That's exactly what the solution shows:

- **Subnet 2 (≥90)** → **223.1.17.0/25** (usable .1–.126)
- **Subnet 1 (≥60)** → **223.1.17.128/26** (usable .129–.190)
- **Subnet 3 (≥12)** → **223.1.17.192/28** (usable .193–.206)

(They listed them as "1, 2, 3" but the allocation order was largest→smallest.)

4) Why these exact network addresses?

Because each subnet must begin on a boundary of its block size:

- /25 boundaries every **128** → 0, 128 (but 0 used for the first /25)
- /26 boundaries every **64** → 0, 64, **128**, 192 (we pick 128, since 0–127 already taken)
- /28 boundaries every **16** → **192**, 208, 224, 240 (192 is the first free boundary)

5) Could other answers work?

Yes—**any three non-overlapping blocks** of sizes /25, /26, /28 **inside 223.1.17.0/24** are valid, as long as they're boundary-aligned and don't overlap. This layout is just the neatest left-to-right packing.

That's it: count needs → choose smallest fitting CIDR → place largest first on proper boundaries. Barakah in tidy subnets. 🌙

give me a detailed and general approach for solving the type of problems I gave you

Perfect question 👌 — because both of your tasks (the “192.i…” departmental subnetting” and the “223.1.17.x/24 splitting”) are just **applications of the same general subnetting/VLSM method**. Let me give you a full **step-by-step strategy** that works for *any* such problem.

🔑 General Approach to Subnetting / VLSM Problems

Step 1 – Understand the requirements

- **Base network**: You’re usually given a starting network (e.g., `192.53.0.0/24` or `223.1.17.0/24`).
- **Departments / Subnets**: Each group has a minimum **host requirement** (e.g., 60, 90, 12). Sometimes they’re given by a formula.
- **Constraints**: All subnets must stay inside the base block, non-overlapping, and aligned.

Step 2 – Calculate the host requirement for each subnet

- Formula:
  For *n* hosts required → find smallest `2^h - 2 ≥ n`.
- Then block size = `2^h` addresses.
- Prefix length = `32 - h`.

👉 Example: Need 90 hosts

- 2^6 − 2 = 62 < 90 ❌
- 2^7 − 2 = 126 ≥ 90 ✅ → requires **/25**

Step 3 – Sort the subnets by size (largest → smallest)

- This prevents fragmentation.
- Always place the biggest block first, then go downwards.
- Think of it like fitting boxes in a container.

Step 4 – Allocate subnets in order

- Each subnet must **start at a boundary** that matches its size.
- Boundaries come from block sizes:
  - /25: multiples of 128
  - /26: multiples of 64
  - /27: multiples of 32
  - /28: multiples of 16
  - etc.
- Place the first (largest) subnet at the lowest available boundary, then move to the next free aligned start.

👉 Example: Inside `223.1.17.0/24`

- /25 (128 addresses) → must start at .0 or .128. Pick **.0/25**.
- /26 (64 addresses) → must start at multiples of 64 → next free = .128/26.
- /28 (16 addresses) → must start at multiples of 16 → next free = .192/28.

Step 5 – Record details for each subnet

For each, write:

- **Network address** (starting point)
- **Prefix (CIDR)** (e.g., /25)
- **Subnet mask** (e.g., 255.255.255.128)
- **Usable range** (network+1 → broadcast−1)
- **Broadcast address**

Step 6 – Double-check

- Do all usable hosts ≥ requirement?
- Do subnets overlap?
- Are they inside the original block?

If all ✅ → done.

📝 Summary Formula

1. Required hosts → smallest `2^h - 2` ≥ hosts
2. Prefix = 32 − h
3. Align to multiples of block size (2^h)
4. Allocate largest first → smallest
5. Write table: Network / Mask / Usable range

✅ That's the **general reusable method**.
Both your earlier problems are direct applications of this:

- First one → formulas for devices, then fit into /24 using VLSM.
- Second one → explicit host needs (60, 90, 12), then carve with /25, /26, /28.

Would you like me to **make a one-page "cheat sheet" diagram** (with block size table, prefix ↔ hosts conversion, alignment rules) that you can keep for exams?

yes

Here's your **one-page VLSM Subnetting Cheat Sheet** 📄

✅ Contains general steps, prefix-to-host conversion table, and alignment rules.

You can download it here:
VLSM_Cheat_Sheet.pdf

Would you like me to also add a **visual diagram** (like a block diagram of a /24 being split into /25, /26, /28, etc.) inside the cheat sheet, so it's even easier to memorize?

give the cheet sheet here

Got it 👍 Let me write out the **cheat sheet here in text form** so you can copy-paste or even print directly.

📘 VLSM Subnetting Cheat Sheet

🔹 General Steps

1. **Requirements:** Note the base network and required hosts per subnet.
2. **Hosts → Prefix:** Find smallest block where `2^h − 2 ≥ hosts`. Prefix = `32 − h`.
3. **Sort:** Arrange subnets largest → smallest.
4. **Allocate:** Place each subnet at aligned boundary (block size multiple).
5. **Record:** Network, Prefix, Mask, Usable range, Broadcast.
6. **Verify:** Enough hosts? No overlap? Inside base?

🔹 Common Prefix Sizes

Prefix	Block Size	Usable Hosts	Subnet Mask
/25	128	126	255.255.255.128
/26	64	62	255.255.255.192
/27	32	30	255.255.255.224
/28	16	14	255.255.255.240
/29	8	6	255.255.255.248
/30	4	2	255.255.255.252

🔹 Subnet Alignment Rules

- **/25** subnets start at multiples of 128 → (0, 128).
- **/26** subnets start at multiples of 64 → (0, 64, 128, 192).
- **/27** subnets start at multiples of 32 → (0, 32, 64, 96, 128, …).
- **/28** subnets start at multiples of 16 → (0, 16, 32, 48, …).
- **Rule of Thumb:** Always allocate **largest first** to avoid fragmentation.

⚡ With these 3 things (Steps + Prefix Table + Alignment Rules), you can solve any subnetting/VLSM problem like the ones you showed me.

👉 Do you also want me to add a **visual block diagram** example (like showing how 223.1.17.0/24 splits into /25, /26, /28)? That way it's not just text but a quick memory image.